

Crossbar RRAM Arrays: Selector Device Requirements During Read Operation

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Abstract—Passive crossbar resistive random access memory (RRAM) arrays require select devices with nonlinear I - V characteristics to address the sneak-path problem. Here, we present a systematical analysis to evaluate the performance requirements of select devices during the read operation of RRAM arrays for the proposed one-selector-one-resistor (1S1R) configuration with serially connected selector/storage element. We found high selector current density is critical and the selector nonlinearity (ON/OFF) requirement can be relaxed at present. Different read schemes were analyzed to achieve high read margin and low power consumption. Design optimizations of the sense resistance and the storage elements are also discussed.

Index Terms—Crossbar, read margin, read scheme, resistive random access memory (RRAM), select device, sneak path.

I. INTRODUCTION

RESISTIVE memory with two-terminal storage elements is identified as a promising candidate for the next-generation nonvolatile memory, due to its inherent $4F^2$ cell size, sub-10-nm scaling potential, high speed, low power consumption, potential for 3-D integration, and simplicity in fabrication [1], [2]. However, with continuously scaled nanodevices and electrodes, challenges related with the sneak path leakage and the interconnect series resistance in such a purely passive crossbar architecture become major concerns as they diminish the read margin and limit the maximum array size [3], [4]. To mitigate these problems, several solutions have been proposed to introduce nonlinear I - V characteristics to the memory cells [5]–[20]. For unipolar resistive devices, an extra diode can be connected in series to inhibit reverse conduction through unselected cells [5]. For the more preferable bipolar resistive devices, bipolar nonlinear select devices are needed. To date, selectors based on different mechanisms, such as Schottky barriers [6]–[8], tunnel barriers [9], metal–insulator transition [10]–[12], Zener diodes [13], mixed ionic electronic conduction (MIEC) [14], [15], and punchthrough diodes [16], have been investigated. Recently, the feasibility of complementary resistive switch [17]–[19] and electromechanical diodes [20] has also

been explored. Despite reported progress on selector performance, a series of bottlenecks on large current density, high ON/OFF ratios, fast switching speed, and process reliability remain to be resolved.

There have been several reports on the relationship between the performance of the storage element and the resistive random access memory (RRAM) crossbar array [21]–[27]. However, a systematic analysis on the performance metrics required for select devices was still missing, which poses a significant challenge for RRAM array development given the increased emphasis on select device development and integration. The most recent International Technology Roadmap for Semiconductors (ITRS) report specified performance requirements of the selectors based mainly on operations at the cell level [3]. However, we expect a different set of performance metrics are needed to ensure proper operation at the memory array level. This paper presents a comprehensive numerical study on the read operation of crossbar arrays where each cell consists of a serially connected nonlinear selector along with the storage element, forming the commonly known one-selector-one-resistor (1S1R) structure. The aim is to provide a reference for future select device design and optimization. Specifically, Section II describes the device model and bias schemes used for the simulations. Section III provides predictions of the selector performance metrics at different array sizes to obtain optimal read margins. Results from different read schemes with improved read margins and power consumptions are also discussed, followed by the discussions on the effects of the interconnect resistance, different sensing circuitry, and resistive storage node parameters.

II. DEVICE AND CROSSBAR ARCHITECTURE

A. Crossbar RRAM Arrays

Typically, a crossbar resistive memory array consists of switching cells sandwiched by two sets of parallel electrodes as word- and bit-lines, as shown in Fig. 1. The transition of the cell element between the high-resistance state (HRS) and the low-resistance state (LRS) occurs when proper voltages are applied on the electrodes. This simple design allows minimal cell size, but the interconnected passive network structure also leads to sneak leakage currents that can severely limit the output read margin. Fig. 1 shows the cause of misreading in crossbar architecture due to sneak path currents through unselected cells. During read operation, parasitic leakage paths through unselected cells in the array lead to inaccurate output signal and can prevent proper identification of the HRS from

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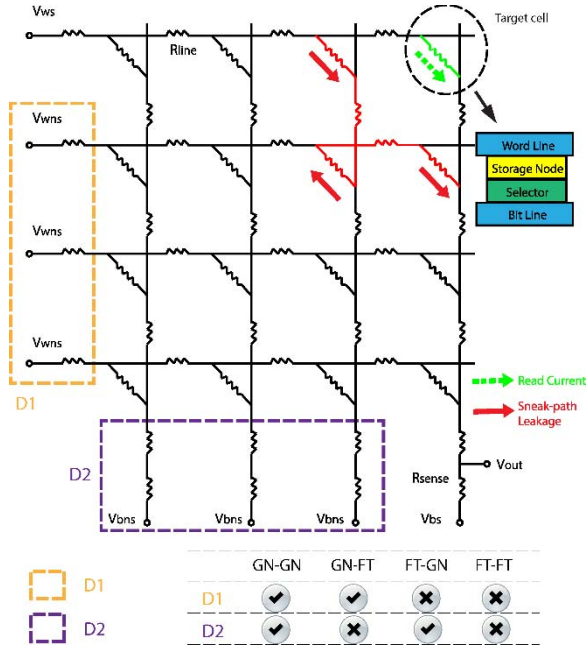


Fig. 1. Schematic of a 4×4 crossbar memory architecture with four different read schemes. Inset: cell structure. Green dashed arrows: the read current through the target cell. Red solid arrows: sneak-path currents through unselected cells may degrade the output voltage. (Box D1/D2) The circuitry inside the boxes will be removed to create a read scheme with floating word- or bit-lines, respectively. GN-GN, GN-FT, FT-GN, and FT-FT stand for different read schemes (see in Section III-B).

the LRS for the target cell. By introducing nonlinear select devices at each crosspoint, it is probable to suppress the sneak path leakage at low voltages and thus improve the read margin.

B. Read Operation

The read operation of crossbar array is achieved by sensing the output voltage, V_{out} . In practice, sense amplifiers are connected with all bit-lines to convert the output current into voltage signal. In our simulation, the peripheral sensing circuit is simplified as a sense resistor R_{sense} . The read margin is defined as

$$RM = \frac{\Delta V_{out}}{V_{read}} = \frac{V_{out}(LRS) - V_{out}(HRS)}{V_{WS}} \quad (1)$$

where V_{WS} is the read voltage applied to the selected word-line, and $V_{out}(LRS)$ and $V_{out}(HRS)$ are the output voltages measured at R_{sense} when reading 1 (LRS) or 0 (HRS). The commonly used ground/ground read scheme was studied first to illustrate the effects of the selector and its performance requirements. In the ground/ground read scheme, to access a specific target cell within the array, we bias the selected word-line to $V_{WS} = V_{dd}$ and the selected bit-lines to $V_{BS} = 0$; while leaving all the unselected word- and bit-lines grounded ($V_{WNS} = V_{BNS} = 0$). This bias mode helps raise the read margin, though it has a major drawback of high power consumption [4]. Several different read schemes will be discussed later in Section III.

Both the data storage pattern in the crossbar RRAM array and the target cell location influence the output voltage swing. Our simulations show that for the commonly used grounding

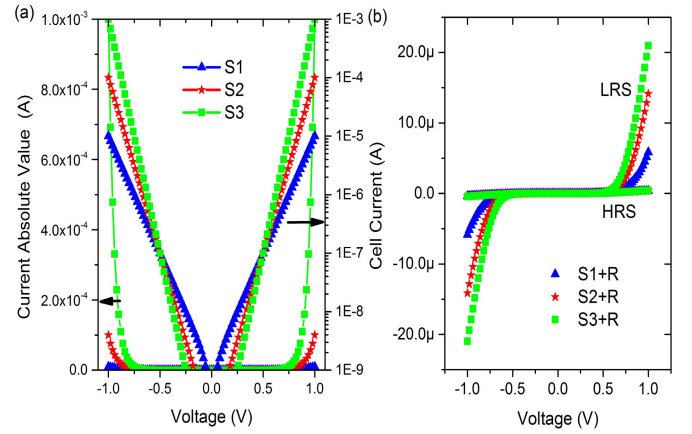


Fig. 2. (a) I - V characteristics of three different selectors (S1-S3) in linear (solid lines) and log (symbols) scales. S1: $k = 10^2$, $I_{sel}(ON) = 10 \mu A$. S2: $k = 10^3$, $I_{sel}(ON) = 100 \mu A$. S3: $k = 10^4$, $I_{sel}(ON) = 1000 \mu A$. (b) I - V characteristics of memory cells serially connected with the selectors and storage elements in (a).

scheme, the worst case for sneak current also results in the worst case for output voltage, and in both cases, the worst case corresponds to the unselected cells being in LRS for reading 1 and 0. Furthermore, due to parasitic interconnect resistance, ΔV_{out} will be smallest for the target cell at the farthest corner from the word/bit-line voltage sources. As a consequence, in this paper, the worst case is chosen as the target cell being at the farthest corner with the unselected cells in LRS.

C. Device Model

In this paper, we mainly focus on bipolar RRAMs, which set/reset processes occur at different bias polarities. Each memory cell in the crossbar array is composed of a nonvolatile storage element serially connected with a symmetric select device, forming the 1S1R structure schematically shown in Fig. 1. The storage node is assumed to have fixed values of R_{ON} and R_{OFF} in the LRS and HRS, respectively, i.e., dynamic switching processes are not studied here since read operation is expected not to disturb the cell states. The I - V characteristics of the select device is modeled by the following hyperbolic sine function:

$$I_{sel} = \gamma \cdot \sinh(\alpha \cdot V) \quad (2)$$

where γ is a conductance parameter, and α represents the nonlinearity of select device. The I - V characteristics of the selector and a memory cell consisting of a storage node serially connected with a selector are shown in Fig. 2(a) and (b). We note that, at moderate voltages the hyperbolic sine function becomes the exponential function used in [27], and also agree reasonably with experimentally observed I - V curves for selector devices based on Schottky diode [8], tunnel barrier [9], MIEC [14], [15], and punchthrough devices [16].

Considering the maximal voltage dropped on an unselected cell is roughly half of the read voltage on the target cell in the ground/ground read scheme, we can further define the ON/OFF ratio for the selector (which describes the nonlinearity of the

TABLE I
SIMULATION PARAMETERS FOR READ OPERATIONS

Parameters	Values
High resistance state (R_{off})	1M Ω
Low resistance state (R_{on})	10k Ω
Sense resistor (R_{sense})	100k Ω
Interconnect resistance between neighboring cells (R_{line})	5 Ω
Voltage at the selected word line (V_{ws})	1V
Voltage at unselected word lines (V_{wns})	0V
Voltage at the selected bit line (V_{bs})	0V
Voltage at the selected bit line (V_{bns})	0V
Selector On/Off ratio (k)	10^4
On-state current of selector ($I_{\text{sel}}(\text{ON})$)	100 μA
Nonlinear factor of selector (α)	18.4207 V $^{-1}$
Selector Conductance Factor (γ)	2×10^{-12} A

selector) as below

$$k \equiv \frac{I_{\text{sel}}(\text{ON})}{I_{\text{sel}}(\text{OFF})} = \frac{I_{\text{sel}}(V_{\text{ws}})}{I_{\text{sel}}(\frac{1}{2}V_{\text{ws}})}. \quad (3)$$

Obviously, ON/OFF ratio is independent of the selector conductance prefactor, γ . For simplicity, the definition above merely describes the nonlinear characteristic of the selector itself, measured at $V_{\text{ws}} = 1$ V, without considering the serially connected storage node. Furthermore, under idealized circumstances with negligible interconnect resistance, negligible sneak path leakages, and negligible voltage divider effect from the select devices [22], the maximum read margin is obtained by setting the sense resistor to be

$$R_{\text{sense}} = \sqrt{R_{\text{ON}} R_{\text{OFF}}}. \quad (4)$$

In Section III, we will first discuss the effects of selector performance parameters on the crossbar array performance. Optimizations by choosing appropriate R_{ON} , R_{OFF} , and R_{sense} will then be investigated. The default parameters used in our simulations are listed in Table I.

III. RESULTS AND DISCUSSION

A. Selector Parameter Dependence

HSPICE simulations were conducted based on the model mentioned in Section II. The read margin of passive crossbar arrays can be calculated numerically for selectors with different characteristics. Fig. 3 summarizes the read margin at different array sizes when selectors with different ON/OFF (nonlinearity) and ON-currents $I_{\text{sel}}(\text{ON})$ are used. Fig. 3(a) shows that for a specific selector ON/OFF ratio of 10^4 , the swing of output voltage remains relatively constant for small arrays but degrades significantly as the crossbar memory array size increases beyond 64×64 . A significant result here is that selectors with larger $I_{\text{sel}}(\text{ON})$ (hence lower effective resistance R_{sel}) leads to much better read margins [e.g., read margin is improved by more than 50% when $I_{\text{sel}}(\text{ON})$ is increased from 10 to 100 μA] due to a smaller voltage divider effect from the selector in the serially connected memory element/selector configuration. Detailed analysis on the HRL and LRS show that the LRS tends to benefit more from the enhanced output

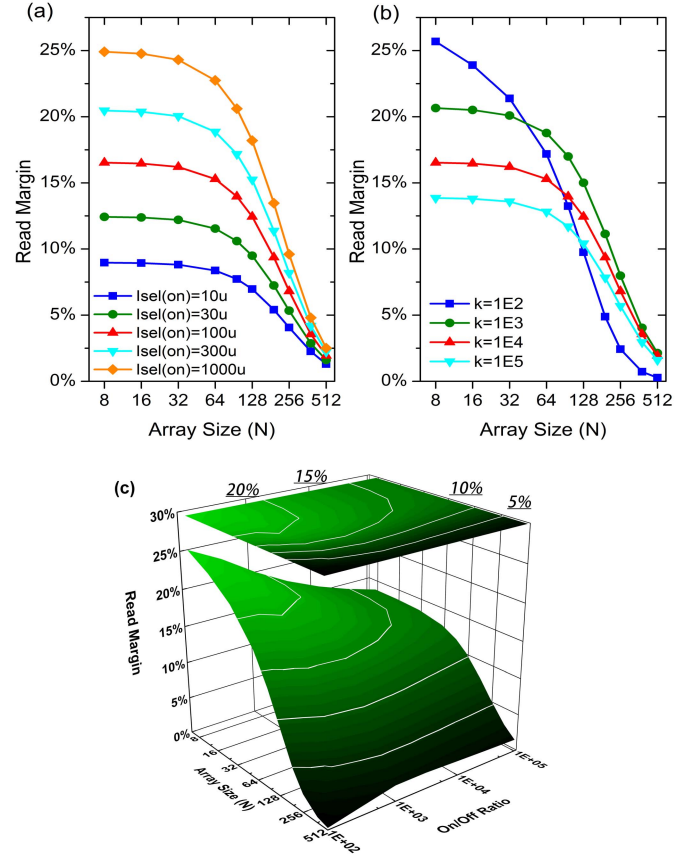


Fig. 3. (a) Read margin as a function of the crossbar array size at different selector ON-currents and a fixed selector ON/OFF ratio $k = 10^4$. (b) Read margin as a function of the crossbar array size at different ON/OFF ratios and a fixed selector ON-current $I_{\text{sel}}(\text{ON}) = 100 \mu\text{A}$. (c) 3-D and 2-D contour plots of the read margin as a function of the array size and the selector ON/OFF ratio. The selector ON-current is kept 100 μA .

voltage compared with the HRS (since the voltage divider effect is more pronounced when the storage element is in the LRS), thus causing a wider read margin. On the other hand, the improvement by increasing $I_{\text{sel}}(\text{ON})$ alone becomes limited in absolute terms as the array size is increased. As shown in Fig. 3(a), for selectors with $k = 10^4$, the read margin drops rapidly below the minimum requirement of 10% when the array exceeds 256×256 due to the sharp increase in sneak currents at very large arrays, regardless of the 100 fold change in selector ON-state current.

Introducing selectors with high ON/OFF ratios can potentially eliminate the sneak-path leakage under low reverse bias, and provide more distinguishable output signals on the sense resistor, particularly for large arrays, as shown in Fig. 3(b). Fig. 3(c) shows the variation of the read margin when the selector ON/OFF ratios are changed [while maintaining the same selector ON-state current $I_{\text{sel}}(\text{ON})$] in crossbar arrays of different sizes. In practice, once the array size and the anticipated read margin are chosen, the allowed parameter range of ON/OFF ratio can be predicted according to the 2-D contour in Fig. 3(c).

One surprising result from Fig. 3(b) and (c) is that a higher selector ON/OFF ratio (i.e., nonlinearity) does not necessarily result in an improved read margin. This observation is

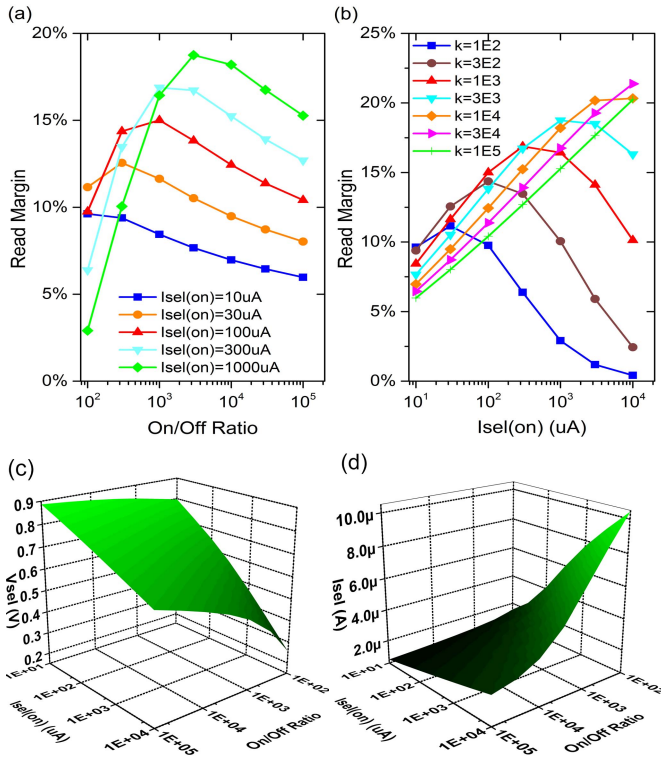


Fig. 4. (a) Effect of selector ON/OFF ratio on read margin. (b) Effect of selector ON-state current on read margin. (c) Bias and (d) current of the selector device in the target cell as functions of selector ON-current and ON/OFF ratios during LRS reading. The size of crossbar array is fixed (128×128) in (a)–(d).

counterintuitive and deserves further investigation. To analyze this effect further, we chose a fixed crossbar array size (128×128) as our model system and systematically studied the effects of selector ON/OFF ratios and ON-currents $I_{sel}(ON)$ on the read margin. The results are shown in Fig. 4. As shown in Fig. 4(a), for a certain $I_{sel}(ON)$, there exists an optimal value of selector nonlinearity and further increasing the selector nonlinearity in fact leads to a decrease (albeit slowly) in read margin. The preferred position of the selector nonlinearity (ON/OFF ratio) also depends on the selector ON-current $I_{sel}(ON)$ and shifts to higher values with increasing peak read margin as the selector becomes more conducting. This nonmonotonic dependence on ON/OFF ratios can be explained by two competing mechanisms: on one hand, increasing the ON/OFF ratio alleviates the sneak-path leakage around the target cell; on the other hand, however, a very large ON/OFF ratio means that a high voltage is needed for the selector to maintain the desired current as a small drop in voltage will severely reduce the current through the selector (and the target cell). As shown in Fig. 4(c) and (d), with increasing ON/OFF ratios, a larger V_{sel} is consumed by the selector even though the target cell current I_{sel} start to decrease due to the increase in selector nonlinearity (ON/OFF) at a given $I_{sel}(ON)$. The reduction of output current accumulated by the sense resistor in turn leads to the reduced read margin at a very large ON/OFF.

At what point the nonlinear behavior plays an adverse role depends on how conductive the selector is, and this problem is particularly pronounced for selectors having a low $I_{sel}(ON)$,

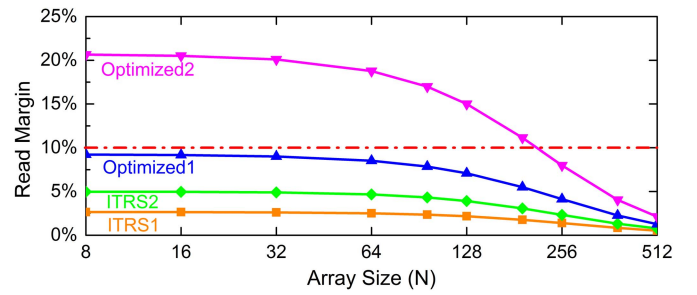


Fig. 5. Read margins in crossbar arrays with different selectors. ITRS1: $k = 10^6$, $I_{sel}(ON) = 1 \mu A$. ITRS2: $k = 10^6$, $I_{sel}(ON) = 4 \mu A$. Optimized1: $k = 10^2$, $I_{sel}(ON) = 4 \mu A$. Optimized2: $k = 10^3$, $I_{sel}(ON) = 100 \mu A$.

as shown in Fig. 4(a) and (b). For the sake of completeness, cases with extremely large ON-state current are also evaluated in Fig. 4(b). In contrast to Fig. 4(a), not all of the curves in Fig. 4(b) can reach their peaks within the simulated parameter range of $I_{sel}(ON)$. A selector with very high ON/OFF ratio (e.g., $>10^4$) does not offer any benefit compared with a selector with a low ON/OFF (e.g., 300) until the selector ON-current is $>200 \mu A$. This analysis suggests obtaining a high $I_{sel}(ON)$ is a key requirement for a nanoscale select device, provided the selection device is of sufficient nonlinearity.

Previous analyses focused mostly on the role of selector ON/OFF while the requirement of selector ON-current was normally relaxed. For example, the benchmark prediction of ITRS suggested the selector should have an ON-state current of $1 \mu A$ at $1 V$ and an ON/OFF ratio as high as 10^6 [3]. Another widely used reference value for selector current density is $1 MA/cm^2$, corresponding to an ON-state current of $4 \mu A$ for a $20 nm \times 20 nm$ crossbar cell. However, as shown in Fig. 5, our analysis shows that with this level of ON-state current, the optimal ON/OFF selector ratio is in fact around 10^2 , and ON/OFF ratios beyond that will actually hurt the array performance. Also shown in Fig. 5 is a further optimized selector with ON/OFF = 10^3 and $I_{sel}(ON) = 100 \mu A$, which produces much higher read margins compared with the ITRS values. Considering an $I_{sel}(ON)$ of $1 MA/cm^2$ is already very challenging to obtain, this paper suggests an urgent need to develop selectors with high ON-state current.

In a typical selector device, two distinct regions can be observed in its I – V characteristics: subthreshold region at low bias with sharp increase [e.g., similar to (2)] in current, and saturation region at high bias with slowly increasing current. For most of the reported electrical selectors that are the focus of this paper [7]–[9], [14]–[16], their finite subthreshold slope limits how fast the selectors can be switched. Thus, the requirements of high $I_{sel}(ON)$ and high ON/OFF are not decoupled but instead end up competing with each other. With a given subthreshold slope and $I_{sel}(OFF)$, a larger ON/OFF means the selector will consume a larger voltage to reach the desired $I_{sel}(ON)$ hence the selector will exhibit a more pronounced voltage divider effect. This explains the counterintuitive observation that a high selector ON/OFF actually may lead to a deteriorated read margin. To resolve the dilemma, novel selector devices with very sharp subthreshold slopes will be highly desired [10]–[12], [20].

Additionally, characterizing a selector device with $I_{\text{sel}}(\text{ON})$ measured at the read voltage V_{read} (e.g., 1 V) is not optimal. As shown in Fig. 4(c) and (d), during operation the actual voltage of the selector V_{sel} can be much lower than V_{read} . In other words, the selector device does not need to maintain the exponentially increasing current beyond the actual working bias V_{sel} . A better figure of merit will be the selector subthreshold slope and the saturation current (or saturation voltage). For example, a selector with $k = 10^3$ and saturation current of $1.95 \mu\text{A}$ at 0.72 V will be equivalent to a selector with the same nonlinearity factor and ON-current of $100 \mu\text{A}$ measured at 1 V (Optimized2 in Fig. 5), and provides a read margin of 15% for a 128×128 crossbar array.

B. Optimal Read Configurations

The analysis so far employed the commonly used grounding scheme with grounded unselected word lines and bit lines (GN-GN) during read [4], [22], [26]. In this section, we examine possible performance improvements from optimizing the read schemes, e.g., by floating, grounding or biasing some of the address lines.

Here, GN-FT is referred to the read scheme with grounded unselected word lines and floating unselected bit lines. Similar abbreviations apply to other schemes of FT-GN and FT-FT, as shown in Fig. 1. Additionally, $1/2 \text{ V}$ ($1/3 \text{ V}$) schemes where the unselected word-lines are biased at $1/2$ ($1/3$) V_{read} and the unselected bit-lines are biased at $1/2$ ($2/3$) V_{read} are also considered. To provide clear insight into the different read schemes, we define the sneak-path coefficient as $\theta = I_{\text{sense}}/I_{\text{cell}}$, where I_{sense} , I_{cell} are the currents flowing through the sense resistor and the target memory cell, respectively. Obviously $\theta = 1$ represents the sneak current is negligible. θ larger than unity corresponds to a forward injection of current into the selected bit-line; while θ smaller than unity corresponds to a reverse flow of current into the unselected word-lines. Analyzing the sneak-path coefficient θ thus provides us information on how sneak current affects the read margin in the different read schemes. For a 128×128 crossbar array, the read margins and the sneak-path coefficients for the four different read schemes are simulated with various combinations of selector parameters for the target cell in HRS and LRS, respectively (see in Figs. 6 and 7). Results of the ideal case without sneak currents are also included.

In terms of read margin, it is clear that for any given scheme the read margin will be improved as θ is made closer to 1. Fig. 6 suggests GN-FT and $1/3 \text{ V}$ normally offer the best read margins. The lower read margins in FT-GN and GN-GN can be attributed to the downward current paths to ground along the unselected bit-lines in the FT-GN and GN-GN setups. This leads to an increase in supply current which in turn raises the voltage drop on interconnect series resistance in the selected word-line and results in a reduced effective read voltage across the target cell.

We note the performances of the $1/2 \text{ V}$ and FT-FT schemes strongly depend on the switching function of the select device. The potentials at the unselected word-lines are raised by biasing ($1/2 \text{ V}$) or by the sneak currents through the

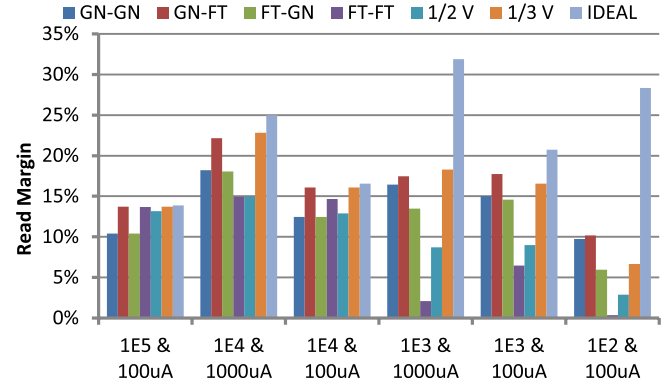


Fig. 6. Comparison of read margins using the six read schemes and in the ideal case without sneak currents. Six different parameter combinations of selector are used. The crossbar array size is fixed at 128×128 .

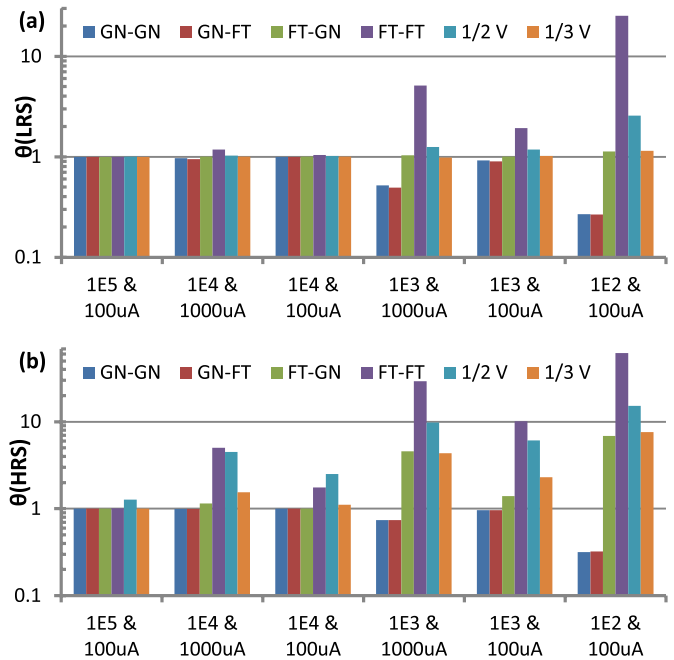


Fig. 7. Sneak-path coefficients for different read schemes within a fixed RRAM array (128×128) at (a) LRS and (b) HRS.

unselected bit-lines (FT-FT). Therefore, at low ON/OFF ratios, large amount of current is injected into the selected bit-line and gives rise to an unrecognizable output voltage swing, as shown in Fig. 7. However, when highly nonlinear selectors are introduced, the sneak path leakage from unselected word-lines will be inhibited significantly and both schemes can exhibit a desirable read margin comparable with that of GN-FT.

Additionally, it remains true that in all configurations an enhanced selector conductance is preferred for better array performance, consistent with earlier conclusions. Furthermore, analyses on the extreme cases illustrate that, when selectors are more ideal, e.g., with $\text{ON/OFF} = 10^5$ and $I_{\text{sel}}(\text{ON}) = 100 \mu\text{A}$, the read margins of the GN-FT, FT-FT, $1/2 \text{ V}$, and $1/3 \text{ V}$ configurations are very similar to the ideal case without any sneak currents (e.g., Fig. 6), but the GN-GN and FT-GN configurations still exhibit lower read margins due to

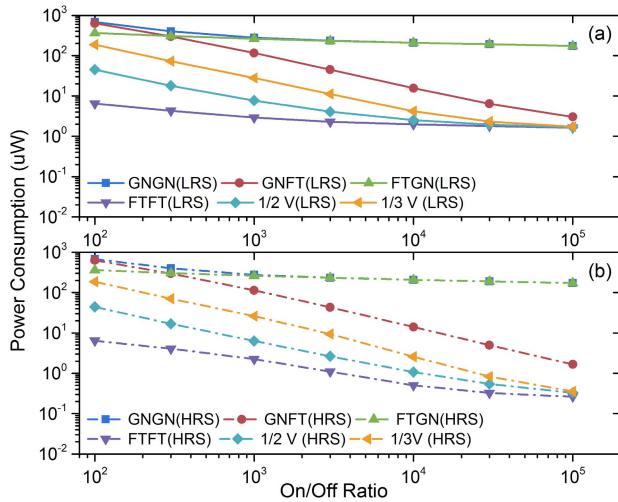


Fig. 8. Overall power consumption of the entire crossbar array (128×128) with different read schemes for reading (a) 1 and (b) 0. The ON-state current of selector is kept at $100 \mu\text{A}$.

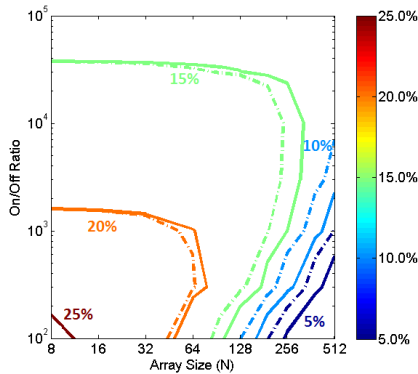


Fig. 9. 2-D contour plots showing the read margin affected by interconnect resistance using the GN-FT scheme. The selector ON-state current is kept at $100 \mu\text{A}$. Dashed line: $R_{\text{line}} = 5 \Omega$. Solid lines: $R_{\text{line}} = 2.5 \Omega$.

downward current paths through the unselected bit-lines to ground in these two configurations.

Power consumption during read operation can be another concern and is analyzed in Fig. 8 for different read schemes. It can be observed that the FT-FT scheme offers by far the lowest power consumption and the GN-GN scheme offers the highest power consumption in general. The $1/2 V$ also offers improved power dissipation since it minimizes the current through the unselected cells due to its symmetric bias on unselected word- and bit-lines. The $1/3 V$ scheme offers another compromise since the power dissipation can be minimized as the selector becomes more nonlinear.

C. Influence of Interconnect Resistance

Fig. 9 shows the read margins influenced by different interconnect resistances using the GN-FT scheme. We note that when the array size is small the read margin is insensitive to the interconnect resistance. However, as the number of cells increases, the interconnect resistance can negatively affect the array performance.

Three factors from interconnect resistance cause the degradation of read margin: 1) voltage dropped on the selected

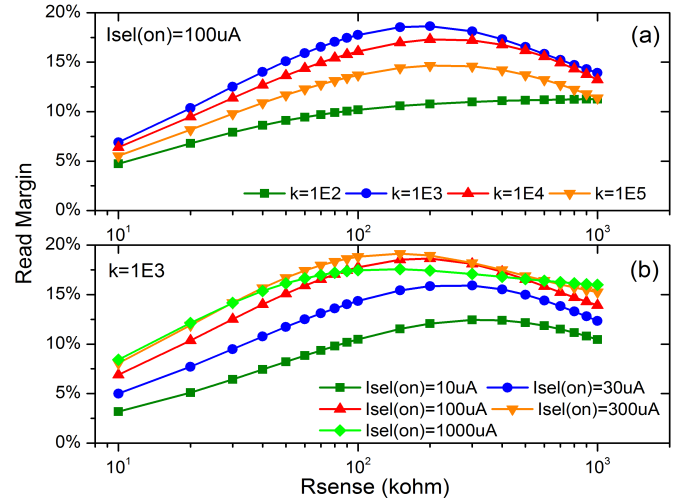


Fig. 10. Read margin's dependence on sense resistor when (a) selector ON-state current $I_{\text{sel}}(\text{ON})$ is fixed ($100 \mu\text{A}$) and (b) selector ON/OFF ratio k is fixed (10^3).

word-line; 2) voltage dropped on the selected bit-line; and 3) the raising or lowering of potential on unselected word-lines that can drive parasitic current into (or out of) the sense resistor. Therefore, even though nonlinear selectors are incorporated in crossbar arrays, reducing the interconnect resistance is still crucial for large array operation. All three effects can be verified by analyzing the potentials on the word- and bit-lines at different nodes in the array during simulation.

As can be observed from Fig. 9, for a fixed array dimension, electrode lines with lower R_{line} allows a larger range of selector ON/OFF ratios for a given read margin. Equivalently, if a selector is already chosen, the maximum array size can be improved with more conducting electrodes for a given read margin.

D. Optimal Sense Resistance

According to the simplified model in [22], the maximum read margin is achieved with a sense resistor equal to the geometric mean of R_{ON} and R_{OFF} , i.e., $R_{\text{sense}} = \sqrt{R_{\text{ON}} R_{\text{OFF}}}$. Nevertheless, it needs to be examined what the optimal sense resistor values is when the select device and interconnect resistances are included. The impacts of sense resistor on the read margin is shown in Fig. 10(a) and (b), while keeping the selector $I_{\text{sel}}(\text{ON})$ and ON/OFF ratio fixed. The results indicate that within a wide range of selector parameter space, the optimal value of sense resistor, $R_{\text{sense_op}}$, remains close to the geometric mean of R_{ON} and R_{OFF} . The small shift of $R_{\text{sense_op}}$ can be understood by the increase of equivalent cell resistance due to the serially connected selector or electrode resistance. The more conducting the selector is, the smaller the shift is. Since the read margin varies slowly with the sense resistor, possible tradeoff adjustments on the sensing circuitry are feasible when other practical parameters need to be optimized, such as power consumption and fan-out.

E. Storage Node Parameters

Below, we discuss how the storage node parameters affect the read margin in the selector/storage node configuration.

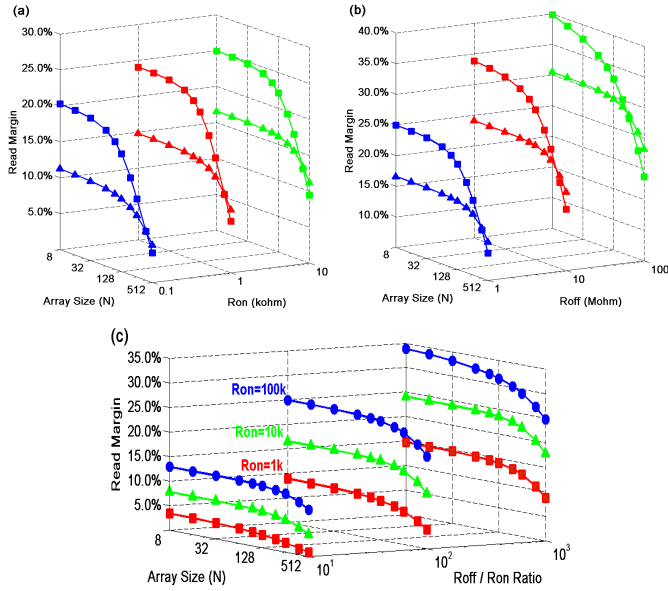


Fig. 11. Relationship between the read margin and the storage node parameters. (a) With fixed $R_{OFF} = 1 \text{ M}\Omega$. (b) With fixed $R_{ON} = 10 \text{ k}\Omega$. (c) With varying R_{OFF}/R_{ON} ratios. In both (a) and (b), upper curves marked by squares correspond to Selector1 ($k = 10^4$, $I_{sel}(ON) = 100 \text{ }\mu\text{A}$); lower curves marked by triangles correspond to Selector2 ($k = 10^4$, $I_{sel}(ON) = 1000 \text{ }\mu\text{A}$). Selector1 is simulated in (c). The sense resistor in (a)–(c) is set to $(R_{ON}R_{FF})^{1/2}$.

In particular, since we only focus on the read operation, the storage node can be modeled as a resistor with resistance value of R_{ON} or R_{OFF} , at the HRS or LRS, respectively.

Fig. 11 shows the relationship between the storage node resistance values and the read margin. For a given selector, in Fig. 11(a), we observe that lowering R_{ON} does not necessarily lead to improved read margin, even though the R_{OFF}/R_{ON} ratio is improved. This is because reducing R_{ON} in fact aggravates the voltage divider effect from the serially connected nonlinear select devices and the finite interconnect resistances. On the other hand, an opposite trend is found in Fig. 11(b), which shows that increasing R_{OFF} leads to increased read margin at the same R_{OFF}/R_{ON} ratio. This trend is further verified by examining the read margin as a function of storage node ON/OFF ratio for different R_{ON} values. As shown in Fig. 11(c), the read margin generally increases as the R_{OFF}/R_{ON} ratio is increased, but is always higher for higher R_{ON} at the same R_{OFF}/R_{ON} ratio due to the voltage divider effect.

These results suggest that if the device can only provide a limited R_{OFF}/R_{ON} ratio, a desirable solution to mitigating the output signal degradation is to have high values of both R_{ON} and R_{OFF} instead of low values. However, in practical designs, this requirement needs to be balanced with other demands such as speed requirements.

IV. CONCLUSION

The read operations of RRAM crossbar array have been systematically studied for a proposed 1S1R configuration consisting of memory cells with serially connected select device and storage node. The read margin was found to closely depend on the nonlinear conduction of the select device. Increasing the crossbar array size demands high values of

not only selector ON/OFF ratio (nonlinearity), but also high selector current density especially at low bias, and previously benchmarked selector ON-current values seem too low for practical applications. The GN-FT and $1/3 \text{ V}$ schemes were found to offer better read margin and lower operation power compared with the conventional GN-GN scheme. If highly nonlinear selectors can be obtained, the FT-FT and $1/2 \text{ V}$ schemes are more desirable for ultralow power application. The optimal sense resistor value was found to be still around the geometric mean of R_{ON} and R_{OFF} but only affects the read margin slowly. The parameters of the storage node device should be optimized by considering the inherent voltage divider effect from the serially connected selector. These results provide a theoretical guidance for the design and optimization of RRAM devices and circuits.

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